

Chapter 4: Random variables

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Defining random variables

Random variables

- A *random variable(RV)* is a numeric quantity whose value depends on the outcome of a random event
 - We use a capital letter, like X , to denote a random variable
 - The values of a random variable are denoted with a lowercase letter, in this case x
 - For example, $P(X = x)$
- There are two types of random variables:
 - *Discrete random variables* often take only integer values
 - Example: Number of credit hours, Difference in number of credit hours this term vs last
 - *Continuous random variables* take real (decimal) values
 - Example: Cost of books this term, Difference in cost of books this term vs last



Discrete random variable

Let a random variable X denote the number of heads in two coin tosses. Find the probability distribution of X .

- Probability distribution function:

$$f(x) = \begin{cases} 1/4, & x = 0 \\ 1/2, & x = 1 \\ 1/4, & x = 2 \\ 0, & \text{o.w.} \end{cases}$$

- Probability distribution table:

x	0	1	2	o.w.
$f(x)$	1/4	1/2	1/4	0

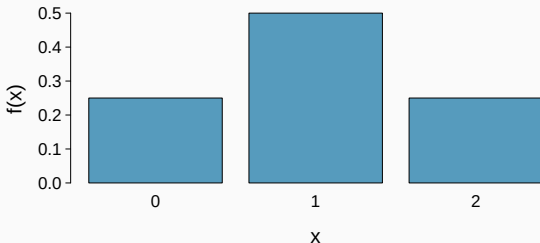


Probability mass function(pmf), $f(x)$

$$f(x) = \begin{cases} P(X = x_i), & x = x_i \ (i = 1, \dots, k) \\ 0, & \text{o.w.} \end{cases}$$

Properties of probability mass function

- $0 \leq f(x) \leq 1, \quad \sum_{i=1}^k f(x_i) = 1.$
- $P(a < X < b) = \sum_{a < x < b} f(x).$



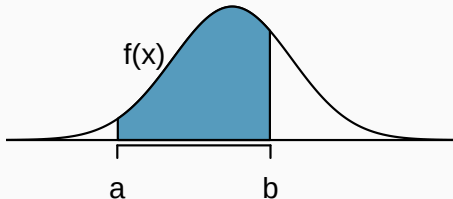
Continuous random variable

Probability density function(pdf), $f(x)$

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

Properties of probability density function

- $f(x) \geq 0$, $\int_{-\infty}^{\infty} f(x)dx = 1$.
- $P(a < X < b) = P(a \leq X \leq b) = \int_a^b f(x)dx$.



Practice

Consider a continuous RV X with a probability density function

$$f(x) = \begin{cases} cx(1-x), & 0 \leq x \leq 1 \\ 0, & \text{o.w.} \end{cases}, \text{ where } c \text{ is a constant.}$$

- Find a constant c so that $f(x)$ is a probability density function.

$$1 = \int_{-\infty}^{\infty} f(x)dx = \int_0^1 cx(1-x)dx = c \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{c}{6} \implies c = 6$$

- Find $P(X \leq 1/2)$.

$$P(X \leq 1/2) = \int_0^{1/2} 6x(1-x)dx = 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^{1/2} = \frac{1}{2}$$



Expectation and variance

If you repeat tossing two coins, how many heads would you expect to appear on average?



Expectation

We are often interested in *average* outcome of a random variable.

Expectation(mean) of a RV X

$$\mu = E(X) = \begin{cases} \sum_{i=1}^k x_i f(x_i) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Expectation of a function of RV $g(X)$

$$E(g(X)) = \begin{cases} \sum_{i=1}^k g(x_i) f(x_i) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} g(x) f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$



Practice

Let a random variable X denote the number of heads in two coin tosses. Find the expectation of X .

The probability distribution of X is given as

x	0	1	2
$f(x)$	1/4	2/4	1/4

- $E(X) = \sum_x xf(x) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{2}{4} + 2 \cdot \frac{1}{4} = 1$

Find the expectation of $Y = (X - 1)^2$.

- $$\begin{aligned} E(Y) &= E[(X - 1)^2] = \sum_x (x - 1)^2 f(x) \\ &= (0 - 1)^2 \cdot \frac{1}{4} + (1 - 1)^2 \cdot \frac{2}{4} + (2 - 1)^2 \cdot \frac{1}{4} = \frac{1}{2} \end{aligned}$$



Variance and standard deviation

We are also often interested in the *variability* of a random variable.

Variance and standard deviation of a RV X

$$\sigma^2 = \text{Var}(X) = E[(X - \mu)^2] = \begin{cases} \sum_{i=1}^k (x_i - \mu)^2 f(x_i) & X \text{ discrete} \\ \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx & X \text{ continuous} \end{cases}$$

$$\sigma = \text{SD}(X) = \sqrt{\text{Var}(X)}$$

Computing variance

$$\text{Var}(X) = E(X^2) - E(X)^2$$

Note:

$$\text{Var}(X) = E[(X - \mu)^2] = E(X^2 - 2\mu X + \mu^2) = E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - E(X)^2$$



Practice

Tony is waiting for a bus. Suppose that waiting time X (hours) has the following a probability density function $f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{o.w.} \end{cases}$

- Find $E(X)$:

$$E(X) = \int_0^1 x \, dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}, \quad E(X^2) = \int_0^1 x^2 \, dx = \frac{1}{3}$$

- Find $\text{Var}(X)$:

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$



Properties of expectation and variance

Let a and b be constants(not random).

Property of expectation

$$E(aX + b) = aE(X) + b$$

Properties of variance

- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- $\text{Var}(X) \geq 0$

Note: If X is a continuous random variable(discrete X can be shown similarly),

$$E(aX + b) = \int_{-\infty}^{\infty} (ax + b)f(x)dx = a \int_{-\infty}^{\infty} xf(x)dx + b \int_{-\infty}^{\infty} f(x)dx = aE(X) + b$$

$$\begin{aligned}\text{Var}(aX + b) &= E(aX + b - E(aX + b))^2 = E(aX + b - aE(X) - b)^2 \\ &= a^2 E(X - E(X))^2 = a^2 \text{Var}(X)\end{aligned}$$



Practice

Tony is waiting for a bus. Suppose that waiting time X (hours) has the following a probability density function $f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{o.w.} \end{cases}$

Let $Y = 2X + 1$.

- Find $E(Y)$:

$$E(Y) = E(2X + 1) = 2E(X) + 1 = 2 \times \frac{1}{2} + 1 = 2$$

- Find $\text{Var}(Y)$:

$$\text{Var}(Y) = \text{Var}(2X + 1) = 4 \text{Var}(X) = 4 \times \frac{1}{12} = \frac{1}{3}$$



Joint distributions

Recall: Conditional probability

The conditional probability of the outcome of interest A given condition B is calculated as

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} \quad \begin{array}{l} \leftarrow \text{joint probability} \\ \leftarrow \text{marginal probability} \end{array}$$

Joint probability: Probability for two events(variables).

Marginal probability: Probability for a single event(variable).



Joint probability distribution

Joint pmf of two **discrete** RVs X and Y

$$f(x, y) = P(X = x, Y = y)$$

- $0 \leq f(x, y) \leq 1, \sum_{\forall x} \sum_{\forall y} f(x, y) = 1.$
- $P(a < X \leq b, c < Y \leq d) = \sum_{a < x \leq b} \sum_{c < y \leq d} f(x, y)$

Joint pdf of two **continuous** RVs X and Y

$$\int_a^b \int_c^d f(x, y) dy dx = P(a < X < b, c < Y < d)$$

- $0 \leq f(x, y), \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1.$



Marginal probability distribution

Marginal pmf of a *discrete* RV

$$f_X(x) = \sum_{\forall y} f(x, y), \quad f_Y(y) = \sum_{\forall x} f(x, y)$$

Marginal pdf of a *continuous* RV

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$



Practice

In a certain suburban area, each household reported the number of cars X and the number of television sets Y that they owned. Joint pmf of X and Y is given below.

$X \backslash Y$	1	2	3	4
1	0.1	0.0	0.1	0.0
2	0.3	0.0	0.1	0.2
3	0.0	0.2	0.0	0.0

- Find the marginal pmf of X :

x	1	2	3
$f_X(x)$	0.2	0.6	0.2

- Find the marginal pmf of Y :

y	1	2	3	4
$f_Y(y)$	0.4	0.2	0.2	0.2



Independence of two variables

Recall: Independence of two events

Two events A and B are independent if

$$P(A \text{ and } B) = P(A) \times P(B).$$

Independence of two random variables

X and Y are (mutually) independent if

$$f(x, y) = f_X(x)f_Y(y).$$



Practice

Joint pdf of X and Y is given by

$$f(x, y) = \begin{cases} \frac{3x - y}{9} & 1 < x < 3, 1 < y < 2 \\ 0, & \text{o.w.} \end{cases}$$

Determine whether X and Y are independent.

$$f_X(x) = \int_1^2 \frac{3x - y}{9} dy = \frac{x}{3} - \frac{1}{6}, \quad 1 < x < 3$$

$$f_Y(y) = \int_1^3 \frac{3x - y}{9} dx = \frac{4}{3} - \frac{2y}{9}, \quad 1 < y < 2$$

$$f(x, y) \neq f_X(x)f_Y(y) \implies X \text{ and } Y \text{ are not independent.}$$



Covariance

Expectation of a function of RVs $g(X, Y)$

$$E(g(X, Y)) = \begin{cases} \sum_{\forall x} \sum_{\forall y} g(x, y) f(x, y) & \text{if } X, Y \text{ discrete} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy & \text{if } X, Y \text{ continuous} \end{cases}$$

Covariance measures the joint variability of two random variables.

Covariance

Covariance of two random variables X and Y is

$$\begin{aligned} \text{Cov}(X, Y) &= E[(X - E(X))(Y - E(Y))] \\ &= E(XY) - E(X)E(Y). \end{aligned}$$

Note: If X and Y are independent, $\text{Cov}(X, Y) = 0$, but not vice versa.



Practice

Joint and marginal pmf of X and Y are given below.

$X \backslash Y$	1	2	3	4	$f(x)$
1	0.1	0.0	0.1	0.0	0.2
2	0.3	0.0	0.1	0.2	0.6
3	0.0	0.2	0.0	0.0	0.2
$f(y)$	0.4	0.2	0.2	0.2	1.0

- Find $E(X)$ and $E(Y)$:
 $E(X) = \sum_x xf(x) = 2$
 $E(Y) = \sum_y yf(y) = 2.2$
- Find $E(XY)$:
 $E(XY) = \sum_x \sum_y xyf(x, y)$
 $= 4.4$

- Find $Cov(X, Y) := E(XY) - E(X)E(Y) = 4.4 - 2 \times 2.2 = 0$

Note: Are X and Y independent? $f(1, 1) = 0.1 \neq f_X(1)f_Y(1) = 0.08 \implies$ not independent.



Correlation

Correlation of two random variables X and Y is

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

Properties of correlation

- $-1 \leq \rho(X, Y) \leq 1$
- $\rho(X, Y) > 0$: positive linear relationship between X and Y .
- $\rho(X, Y) < 0$: negative linear relationship between X and Y .
- $\rho(X, Y) = 0$: no linear relationship between X and Y .

Linear combinations

A *linear combination* of random variables X and Y is given by

$$aX + bY$$

where a and b are some fixed constants.

The average value of a linear combination of RVs is given by

Expectation of a linear combination of two RVs

$$E(aX + bY) = a \times E(X) + b \times E(Y)$$

The variance of a linear combination of two *independent* RVs is

Variance of a linear combination of two RVs

$$\text{Var}(aX + bY) = a^2 \times \text{Var}(X) + b^2 \times \text{Var}(Y)$$



Practice

On average a notebook costs \$10 and a pen costs \$15. You plan to buy 5 notebooks (all at the same random price X) and 4 pens (all at the same random price Y). What is the expected total cost $E(5X + 4Y)$?

$$E(5X + 4Y) = 5E(X) + 4E(Y) = 5 \times 10 + 4 \times 15 = 110$$

The standard deviation of the price is \$1.5 for a notebook and \$2 for a pen. Assuming independence between X and Y , what is the standard deviation of the total cost $SD(5X + 4Y)$?

$$Var(5X + 4Y) = 25 Var(X) + 16 Var(Y) = 25 \times 1.5^2 + 16 \times 2^2 = 120.25$$

$$SD(5X + 4Y) = \sqrt{120.25} \approx 10.97$$



Simplifying random variables

$2X = X + X$ can be understood as a linear combination of two random variables X and X . Thus,

$$E(X + X) = E(X) + E(X) = 2E(X),$$

but

$$\text{Var}(X + X) \neq \text{Var}(X) + \text{Var}(X),$$

because X and X themselves are not independent. Instead,

$$\text{Var}(X + X) = \text{Var}(2X) = 2^2 \text{Var}(X) = 4 \text{Var}(X).$$



Variance of Linear Combinations: General Case

The variance formula for $aX + bY$ assumed independence. In general, covariance must be included.

Variance of a linear combination (general)

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2 \text{Cov}(X, Y)$$

$$\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) \pm 2ab \text{Cov}(X, Y)$$

Note: When X and Y are independent, $\text{Cov}(X, Y) = 0$, and the formula reduces to

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y),$$

which is the formula from the previous slide.



Exercises in OpenIntro Statistics 4th ed.

- **Conditional probability:**

- Exercise 3.22, 3.42.

- **Random variables:**

- Exercise 3.34, 3.44 (a) and (b), 3.47(or 3.45)

1. Suppose that X and Y have a joint probability density function

$$f(x, y) = \begin{cases} \frac{3}{2}y^2, & 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0, & \text{o.w.} \end{cases}$$

1. Find the marginal pdf of Y , $f_Y(y)$.
2. Find $E(Y)$ and $Var(Y)$.



2. Suppose that X and Y have a joint probability mass function

$$f(x, y) = \begin{cases} \frac{1}{12}(x + y), & x = 0, 2, \text{ and } y = 0, 1, 2, \\ 0, & \text{o.w.} \end{cases}$$

1. Find the marginal pmf of X , $f_X(x)$. (Hint: Use a table in slide 15).
2. Find the marginal pmf of Y , $f_Y(y)$.
3. Find $E(X)$ and $E(Y)$.
4. Find $E(XY)$ and $\text{Cov}(X, Y)$.



3. Suppose that X and Y have a joint probability density function

$$f(x, y) = \begin{cases} e^{-x-y}, & x > 0, y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

1. Find the marginal pdfs $f_X(x)$ and $f_Y(y)$.
2. Find $E(X)$ and $E(Y)$.
3. Find $E(XY)$ and $\text{Cov}(X, Y)$.
4. Find $P(X < Y)$.

